

Compressive creep behavior of load transfer component for variable-stiffness FRP cable anchoring system

Jingyang Zhou, Haorui Xiang, Xin Wang*, Leizhi Zhang, Xinquan Chang, Qiwei Xu, Zhishen Wu*

Key Laboratory of C & PC Structures Ministry of Education, National and Local Unified

Engineering Research Center for Basalt Fiber Production and Application Technology, Southeast

University, Nanjing, 211189, China

*Corresponding author's e-mail: xinwang@seu.edu.cn

Abstract

The static compressive property of modified resin-based load transfer components (LTCs) for cable anchoring systems was experimentally determined by considering the modified materials, volume fraction (VF) and hybrid ratio respectively. Five stress levels were applied to evaluate the creep behavior over 1000 hours. The results revealed that vacuuming increased compressive strength by 18.0% to 113.3% and elastic modulus by 22.5% to 48.0%. These properties of single glass microfiber-modified resin (SFMR) and single quartz sand-modified resin (SSMR) show a linear relationship with an adjusted R^2 value exceeding 0.98. All specimens displayed a two-stage creep development, with the addition of sand having a more positive impact on reducing creep strain in SSMR compared to SFMR. Adding fibers reduced the 1000-hour creep coefficient while adding sand had a slight effect. The creep coefficients of SFMR and SSMR rapidly increased when stress levels reached $0.7f_u$ (f_u =ultimate strength) and $0.5f_u$ respectively. Excessive stress levels and VFs reduced strength retention after creep. Both the Findley model and logarithmic model can reasonably forecast million-hour creep behavior. To achieve a creep coefficient within 10%, maximum stress levels for plain resin, SFMR, and SSMR should be set at $0.5f_u$, $0.8f_u$, and $0.5f_u$ respectively.

23 **Keywords**

24 Fiber-reinforced polymer (FRP); Load transfer component (LTC); Modified resin; Compressive
25 behavior; Creep; Prediction

26 **1. Introduction**

27 The lightweight, large span and material-saving properties of cable structures make them widely
28 used in various fields such as cable-stayed bridges and stadiums [1, 2]. Steel cables are still dominant
29 due to their easy anchoring, high stiffness and ductility, and low price [3, 4]. However, steel cables
30 inevitably experience a decline in bearing capacity and service life due to growing self-weight stress
31 and corrosion degeneration [5]. Fiber-reinforced polymers (FRPs) possess outstanding features such
32 as high strength, lightweight, and resistance to corrosion and fatigue [6, 7], making them an ideal
33 candidate for replacing steel cables [8-10].

34 However, FRP cables exhibit viscoelastic behavior due to the presence of a resin matrix [11,
35 12], while resin-based load transfer components (LTCs) are commonly used in cable anchoring
36 systems (CASs) [13-15], which also impart viscoelastic properties to the CASs. Consequently, both
37 FRP cables and LTCs inevitably undergo creep under sustained stress, leading to long-term loss of
38 prestress and increased structural deformation [16]. The creep refers to the material's strain increase
39 under constant stress over time. The high levels of creep stress can further result in creep rupture of
40 FRP cables and anchoring failure, posing significant threats to structural safety. Therefore, it is crucial
41 to focus on evaluating the long-term creep behavior of LTCs and FRP cables.

42 Currently, there is limited research on the creep behavior of LTCs compared to studies on the
43 material itself, which is not used for anchoring purposes. Zhuge et al. [17] found that under axial
44 compressive stresses of 50 and 250 MPa with lateral restraint, the creep strain of epoxy resin-based
45 LTC reached 0.004 and 0.038 respectively. Li et al. [18] discovered that adding long basalt fibers or

46 talcum powder increased the tensile creep behavior of epoxy resin when subjected to creep stresses
 47 at 0.4 and $0.6f_u$ (f_u = ultimate strength). Drozdov [19] studied polypropylene's creep strain under
 48 tensile stresses ranging from 15 to 28 MPa, observing a significant increase in creep deformation over
 49 time at high-stress levels while experiencing minimal changes at low-stress levels. In summary,
 50 multiple factors such as material type, stress level, and temperature [20] can influence the creep
 51 behavior of LTCs.

52 The creep behavior of composite materials, such as FRP, can be categorized into three types
 53 according to different stress levels [21], as illustrated in Fig. 1. When subjected to medium or low
 54 stress, the FRPs exhibit a two-stage creep phenomenon characterized by an initial rapid strain growth
 55 followed by a subsequent stage of strain stability. However, when exposed to high stress, the materials
 56 exhibit an additional third stage characterized by a rapid increase in strain and eventual creep rupture.
 57 Creep rupture only occurs within the first stage under ultra-high-stress conditions. In short, the
 58 aforementioned creep characteristics can serve as a crucial foundation for the analysis and prediction
 59 of creep behavior.

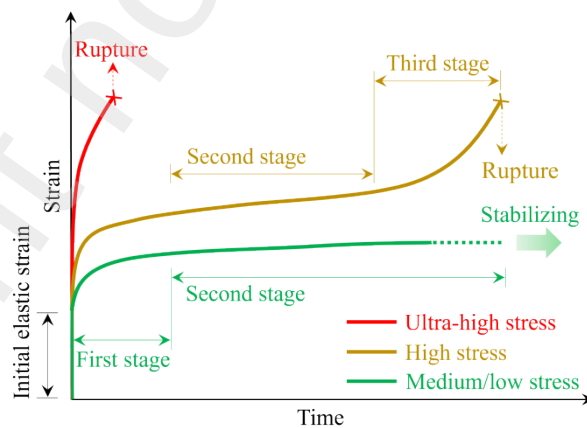


Fig. 1 Creep development under different stress levels

62 Many studies have been conducted to evaluate the creep behavior of FRP tendons. Shi et al. [22]
 63 demonstrated that pretension can increase the creep behavior of basalt FRP (BFRP) tendons, with the
 64 predicted million-hour creep rupture stress increasing from 0.59 to $0.63f_u$. Yang et al. [23] found that

65 after tensile creep tests of 1000 h, carbon FRP (CFRP) tendons had a residual strength that was 4.54%
66 lower but an after-creep elastic modulus that was 6.99% higher compared to static properties. Zou
67 [24] performed a 1000-hour creep test on an aramid FRP (AFRP) tendon with a diameter of 8 mm at
68 stress levels ranging from 0.40 to $0.7f_u$ and determined the million-hour creep coefficient as 13.3%
69 through logarithmic regression analysis. Esmacili [25] discovered that smaller diameter glass FRP
70 (GFRP) tendons were more susceptible to creep rupture than larger ones, and the impact of sustained
71 load on strength reduction was more significant than the combined effect of alkaline solution and
72 high temperature. Mayookh Lal [26] experimentally evaluated the flexural creep behavior of hybrid
73 carbon and glass fibers-reinforced tendons and predicted their deformation over fifty years using the
74 Findley model. In brief, key factors such as stress level, corrosion, temperature, and material type
75 have been comprehensively assessed on FRP tendons.

76 However, research on FRP cables is limited compared to that on FRP tendons. Ai [27] conducted
77 experiments to evaluate the creep behavior of a self-anchored CFRP plate cable system under stress
78 levels ranging from 0.3 to $0.7f_u$ and demonstrated the effectiveness of the logarithmic model in
79 predicting the million-hour creep coefficient. Peng et al. [28] proposed a multiscale model based on
80 impregnated fiber bundles to predict the creep behavior of FRP cables, which showed that the creep
81 rupture time of FRP cables was similar to that of impregnated fiber bundles under low-stress levels.
82 These studies mainly focus on the creep behavior of FRP tendon or cable itself without considering
83 LTCs' creep effect. Additionally, there has been no extension of specific FRP tendon creep research
84 to include FRP cables composed of those same tendons, reducing the significance of studying tendon's
85 creep behavior to a certain extent. Regarding FRP cables, researchers primarily concentrate on
86 developing large-tonnage FRP CASSs while rarely evaluating their creep behavior further. Therefore,
87 it is necessary to conduct comprehensive studies on LTCs' creeps as well as those related to both FRP

88 tendons and cables based on CASs already verified by anchoring and fatigue tests.

89 The previous study proposed a variable-stiffness FRP CAS with glass microfibers-modified
90 LTCs. Static and fatigue results showed that the CAS successfully anchored $\Phi 7$ -37 high-strength
91 CFRP cables (ultimate load = 4070 kN) and $\Phi 7$ -37 BFRP cable (ultimate load = 1919 kN).
92 Additionally, the CAS for BFRP cables survived 2 million cycles after fatigue loading. To further
93 promote the engineering application of the CAS, it is crucial to evaluate the creep behavior of LTCs.
94 Due to the complex three-directional stressed states of microfibers-modified LTCs, this study first
95 focuses on evaluating the compressive creep behavior, while future studies will investigate tensile
96 and shear creep behaviors. These comprehensive results can help to predict the creep behavior of
97 CAS and ensure long-term anchoring safety.

98 The static compressive behavior of single glass fibers-modified resin (SFMR), single quartz
99 sand-modified resin (SSMR), and hybrid glass fibers-modified resin (HFMR) was evaluated.
100 Different volume fractions (VFs) and hybrid ratios, as well as vacuuming, were considered. The
101 failure mode and mechanism, compressive strength, elastic modulus, load-displacement curve, and
102 fitting formula were analyzed. Based on this analysis, the compressive creep behavior of vacuumed
103 SFMR and SSMR was evaluated through 1000-hour creep tests. The effects of the modified material
104 type, stress level, and VF on compressive creep failure modes, creep strain-time curves, and strength
105 retention were analyzed. Finally, million-hour creep coefficients were determined using four
106 prediction models.

107 **2. Static compressive behavior of LTCs**

108 **2.1. Experimental program**

109 **2.1.1. Materials**

110 The properties of the four types of glass microfibers and single quartz sand, which were selected

111 based on previous studies [13, 29], are demonstrated in Table 1. The fiber number was determined
 112 by fiber length. Epoxy resin was used as the matrix, which had guaranteed tensile strength, shear
 113 strength, and compressive strength values of 30, 10, and 70 MPa respectively.

114 Table 1 Properties of glass fibers and quartz sand

Number	Type	Diameter (μm)	Length (mm)	Tensile strength (MPa)	Elastic modulus (GPa)	Density (g/cm ³)
L25/150	Fiber	8	25/150	2800	72	2.56
L20/900	Fiber	13	20/900	2800	72	2.56
QS	Sand	63-88	/	/	70	2.65

115 2.1.2. Specimens

116 As shown in Table 2, the effect of vacuuming to eliminate voids was taken into consideration.
 117 The unvacuumed SFMR included five VFs, while the vacuumed SFMR contained three VFs. The
 118 absence of VFs of 32% and 40% in the vacuumed SFMR was due to limited effectiveness of
 119 vacuuming on the modified resin with poor fluidity. Besides, the maximum VF of 40% was chosen
 120 based on careful consideration of resin's pouring performance. The vacuumed SSMR included five
 121 VFs, which was served as a control group. Five valid specimens were prepared for each condition.

122 Table 2 Classification of plain resin, SFMR and SSMR

Number	Vacuumping	Modified material	VF (%)	Quantity
PR	No	/	/	5
L150-VF8/16/24/32/40	No	L150	8/16/24/32/40	5/5/5/5/5
L900-VF8/16/24/32/40	No	L900	8/16/24/32/40	5/5/5/5/5
V-PR	Yes	/	/	5
V-L150-VF8/16/24	Yes	L150	8/16/24	5/5/5
V-L900-VF8/16/24	Yes	L900	8/16/24	5/5/5
V-QS-VF8/16/24/32/40	Yes	Sand	8/16/24/32/40	5/5/5/5/5

123 Notes: The letter V stands for vacuuming.

As shown in Table 3, long and short fibers were simultaneously used to create four HFMRs. Each type had four hybrid ratios. The aim was to compare them with SFMR while maintaining a fixed total VF of 40%. However, the high fiber VF in HFMR made it unsuitable for direct pouring due to poor liquidity and limited its use as LTCs. Consequently, only a small number of vacuumed HFMRs were prepared to validate the improvement effect of vacuuming. Further investigation into hybrid modification with relatively low fiber VF is recommended for future studies.

Table 3 Classification of HFMR

Number	Vacuumed	VF (%)	Long fiber	Short fiber	Hybrid ratio	Quantity
L150-L25-R41/32/23/14	No	40%	L150	L25	4:1/3:2/2:3/1:4	5/5/5/5
L150-L20-R41/32/23/14	No	40%	L150	L20	4:1/3:2/2:3/1:4	5/5/5/5
L900-L20-R41/32/23/14	No	40%	L900	L20	4:1/3:2/2:3/1:4	5/5/5/5
L900-L25-R41/32/23/14	No	40%	L900	L25	4:1/3:2/2:3/1:4	5/5/5/5
V-L900-L20-R14	Yes	40%	L900	L20	1:4	5
V-L900-L25-R23/14	Yes	40%	L900	L25	2:3/1:4	5/5

Notes: The letter R denotes the hybrid ratio.

The manufacturing process of SFMR can be summarized as follows: Firstly, the epoxy resin and curing agent were mixed at a mass ratio of 2:1 using a stirrer. Then, fibers were added to the resin and stirred until uniform dispersion was achieved. A vacuum defoamer was used to remove bubbles from the mixture's upper surface under a negative pressure of 0.1 MPa. Finally, the mixtures were poured into compressive molds and cured in a drying oven at 40°C for 7 days. Before formal loading, the upper surface of specimens needs further polishing to achieve a required height of 25 mm according to the standard GB/T 2567-2021 [30]. The manufacturing of SSMR and HFMR was similar to SFMR.

2.1.3. Test setup

The compressive test was conducted by the standards GB/T 2567-2021 [30] and GB/T 1448-

2005 [31]. The experimental setup involved the utilization of a universal machine, as depicted in Fig. 2, with a maximum loading capacity of 10 kN. The compressive speed was set at 2 mm/min. Corresponding sensors were employed to record the load and displacement. The compressive strength and elastic modulus can be calculated using Eqs. (1) and (2).

$$\sigma_c = \frac{4F}{\pi d^2} \quad (1)$$

$$E_c = \frac{\sigma_2 - \sigma_1}{\varepsilon_2 - \varepsilon_1} \quad (2)$$

Where σ_c represents the compressive strength, F denotes the ultimate load, d represents the diameter, E_c is the elastic modulus, σ_1 indicates the compressive stress when the compressive strain ε_1 is 0.0005, and σ_2 stands for the compressive stress when the compressive strain ε_2 is 0.0025.

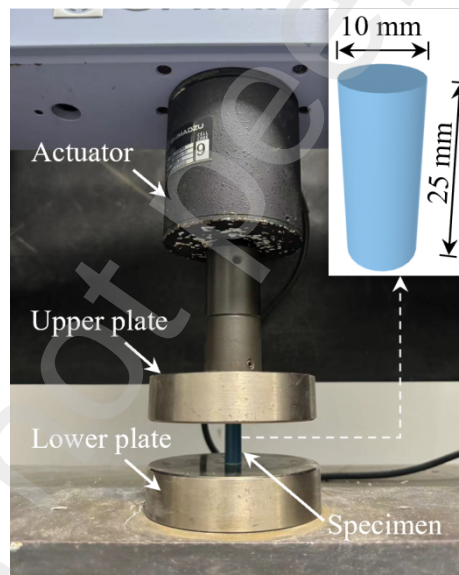


Fig. 2 Compressive test

2.2. Experimental results and discussions

The compressive test results are summarized in Table 4 and Table 5. After vacuuming, the plain resin exhibited an 18.0% increase in compressive strength and a 26.2% increase in elastic modulus. In contrast, the modified resin showed a range of increases in compressive strength from 40.5% to 113.3%, while the elastic modulus also experienced an increase ranging from 22.5% to 48.0%. Additionally, the coefficient of variation for both compressive strength and elastic modulus decreased

159 by less than 5%. In conclusion, vacuuming can be regarded as a viable approach for obtaining accurate
 160 compressive properties. The compressive strength and elastic modulus of the resin modified with a
 161 single L900 were found to be superior compared to those of the resin modified with a single L150.

162 Table 4 Test results for plain resin, SFMR and SSMR

Specimen	Unvacuumed		Vacuumed		Increment (%)	
	σ_c (MPa)	E_c (GPa)	σ_c (MPa)	E_c (GPa)	$\Delta\sigma$	ΔE
PR	74.79 (2.47)	1.41 (2.49)	88.28 (2.01)	1.78 (1.92)	18.0	26.2
L150-VF8	67.01(1.29)	1.73 (3.73)	94.15 (1.20)	2.13 (2.49)	40.5	22.5
L150-VF16	68.45 (2.16)	1.91 (2.43)	99.41 (1.99)	2.44 (2.94)	49.6	27.7
L150-VF24	73.02 (4.73)	2.10 (4.23)	106.78 (3.02)	2.63 (3.43)	46.2	25.2
L150-VF32	75.30 (4.66)	2.24 (8.57)	/	/	/	/
L150-VF40	80.59 (3.63)	2.46 (5.65)	/	/	/	/
L900-VF8	63.29 (4.17)	1.73 (2.19)	97.17 (2.51)	2.23 (2.06)	53.5	28.9
L900-VF16	65.82 (2.91)	1.80 (3.10)	105.05 (2.59)	2.49 (2.68)	59.6	36.7
L900-VF24	70.04 (2.59)	2.07 (2.41)	113.49 (2.72)	2.75 (2.80)	62.0	23.2
L900-VF32	70.86 (4.03)	2.31 (5.92)	/	/	/	/
L900-VF40	71.71 (6.89)	2.35 (5.26)	/	/	/	/
V-QS-VF8	/	/	94.02 (2.20)	1.98 (2.91)	/	/
V-QS-VF16	/	/	99.05 (2.53)	2.21 (3.35)	/	/
V-QS-VF24	/	/	101.61 (2.63)	2.53 (3.54)	/	/
V-QS-VF32	/	/	107.36 (2.96)	2.63 (3.71)	/	/
V-QS-VF40	/	/	115.06 (3.85)	2.83 (3.99)	/	/

163 Note: $\Delta\sigma$ (ΔE) represents the percentage increase in compressive strength (elastic modulus) of the
 164 vacuumed resin compared to that of the unvacuumed resin; the figures in parentheses represent the
 165 coefficient of variation (unit: %).

166 Table 5 Test results for HFMR

Specimen	Unvacuumed		Vacuumed		Increment (%)	
	σ_t (MPa)	E_t (GPa)	σ_t (MPa)	E_t (GPa)	$\Delta\sigma$	ΔE
L150-L25-R41	69.43 (1.60)	2.36 (3.51)	/	/	/	/

L150-L25-R32	69.38 (3.55)	2.32 (3.96)	/	/	/	/
L150-L25-R23	68.20 (3.90)	2.19 (7.65)	/	/	/	/
L150-L25-R14	67.98 (2.66)	2.16 (7.45)	/	/	/	/
L150-L20-R41	66.25 (4.87)	2.46 (6.78)	/	/	/	/
L150-L20-R32	63.51 (4.85)	1.93 (1.85)	/	/	/	/
L150-L20-R23	59.80 (3.10)	1.87 (7.90)	/	/	/	/
L150-L20-R14	57.99 (3.65)	1.88 (3.98)	/	/	/	/
L900-L20-R41	65.81 (2.35)	2.11 (5.12)	/	/	/	/
L900-L20-R32	61.87 (2.53)	2.09 (5.99)	/	/	/	/
L900-L20-R23	56.96 (8.93)	1.83 (6.59)	/	/	/	/
L900-L20-R14	51.71 (4.73)	1.75 (6.64)	110.28 (3.44)	2.59 (2.37)	113.3	48.0
L900-L25-R41	68.13 (2.08)	2.19 (3.87)	/	/	/	/
L900-L25-R32	66.38 (9.03)	2.33 (5.83)	/	/	/	/
L900-L25-R23	66.47 (3.21)	2.34 (5.75)	124.19 (3.12)	3.12 (2.90)	86.8	33.3
L900-L25-R14	66.23 (4.48)	2.23 (5.30)	123.79 (3.01)	3.06 (2.85)	86.9	37.2

2.2.1. Failure mode

The whole buckling was observed in all vacuumed specimens, as shown in Fig. 3, which was mainly caused by a large height-to-diameter ratio of 2.5. The number of diagonal shear cracks increased with the increasing VF of fibers or sand in modified resin indicating that increasing the VF of modified material can enhance the effect of crack deflection and crack path extension [29]. Compared with SSMR, SFMR exhibited a more uniform distribution of cracks and a remarkable slender crack feature, suggesting that fibers had better crack control capacity than sand [13]. This phenomenon can explain the difference in improvement between fibers and sand in subsequent analysis. Local cracking occurred in modified resin with high VFs because the deformation capacity of modified resin decreased with increasing VFs.

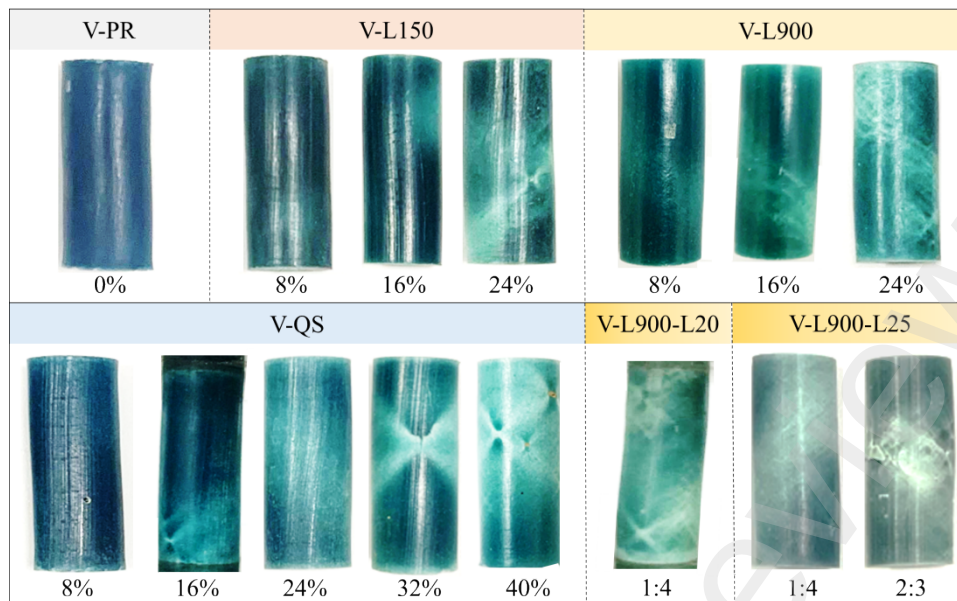


Fig. 3 Compressive failure modes

2.2.2. Load-displacement curve

As presented in Fig. 4, all specimens exhibited a similar three-stage increase (nonlinear, linear, and nonlinear) until reaching the ultimate load. The loading stiffness of SFMR and SSMR gradually increased with an increasing VF of fibers and sand, thereby facilitating the compressive design of a variable stiffness CAS. The post-peak bearing capacity of SSMR showed a continuous decrease; in contrast, SFMR initially declined and then rose, indicating that the crack limiting capacity of the fibers was superior due to their large length-to-diameter ratio. Notably, there was only a slight decrease observed for HFMR. The reason for this is that the fiber VF was 40%, and shorter fibers can achieve a higher fiber count, thereby facilitating a continuous fiber pull-out effect and limiting load reduction. A similar trend has been reported in a previous study [29].

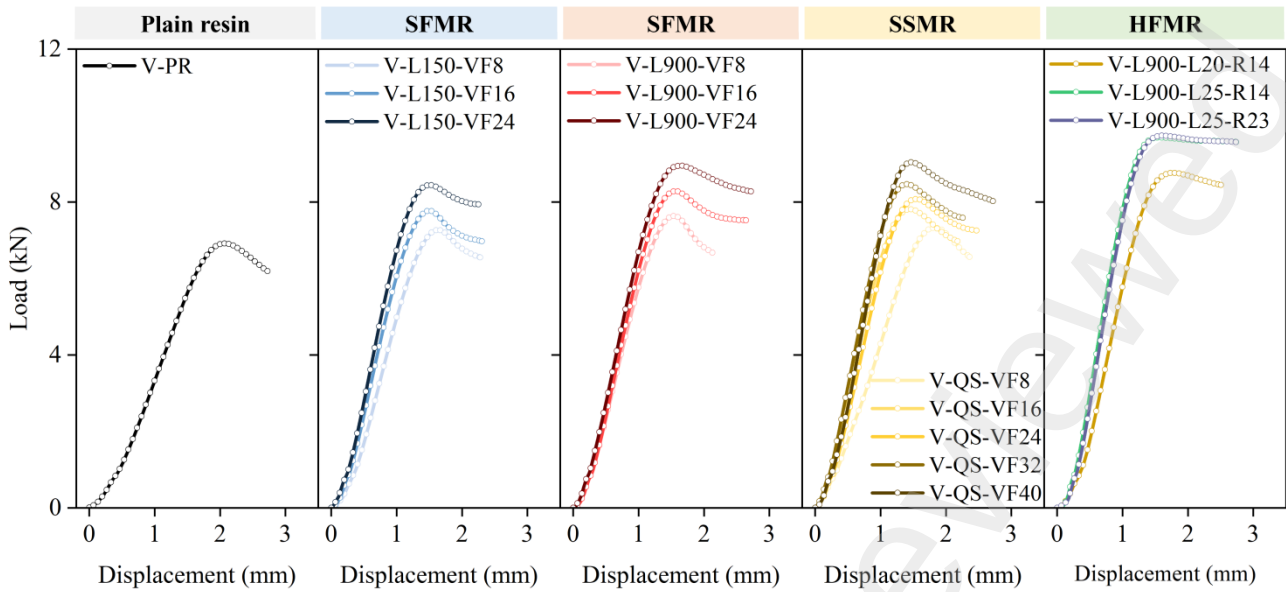


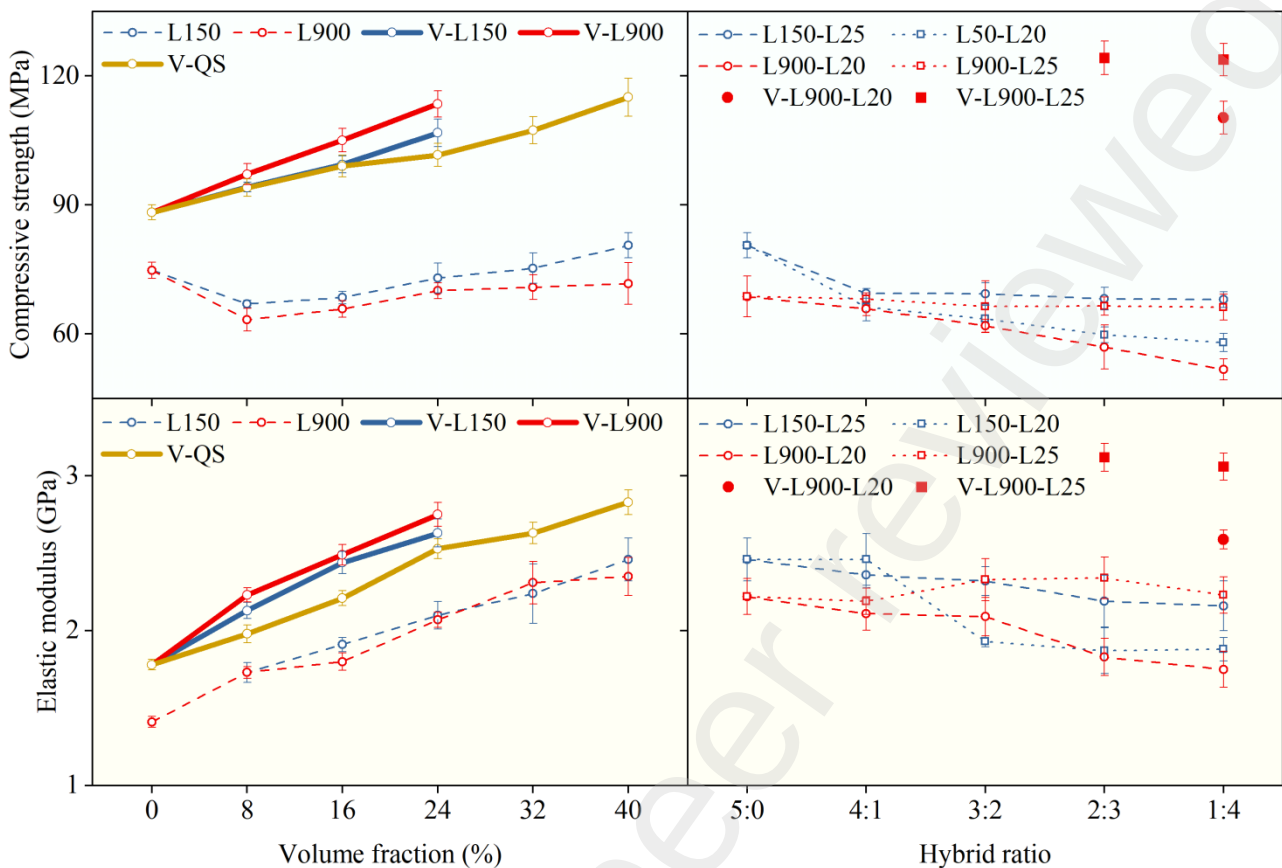
Fig. 4 Compressive load-displacement curves

2.2.3. Compressive strength and elastic modulus

The relationship between compressive strength and VF showed a significant increase after vacuuming, as shown in Fig. 5. This can be attributed to the effective elimination of voids, which prevents crack formation in initial void defects. The compressive strength of SFMR exceeded that of SSMR after vacuuming due to their smaller specific surface area causing sand debonding. In contrast, the unvacuumed HFMR exhibited a gradual decrease in compressive strength with an increasing hybrid ratio. This is because a higher hybrid ratio leads to more short fibers, resulting in poor pouring performance and additional voids; previous studies have also confirmed that long fibers contribute positively to increased compressive strength while short fibers enhance ductility [29].

The elastic modulus of SFMR showed a strong correlation with the fiber VF, regardless of vacuuming. A similar trend was observed in SSMR, indicating comparable enhancing effects of both fibers and sand. It was worth noting that the elastic modulus values for SFMR significantly increased when vacuuming was applied due to reduced voids. Overall, the elastic modulus decreased with an increasing hybrid ratio, except for a few anomalous values; vacuuming also enhanced the hybrid modification effect. In summary, voids had a pronounced impact on compressive strength and elastic

206 modulus and could be effectively eliminated through vacuuming.

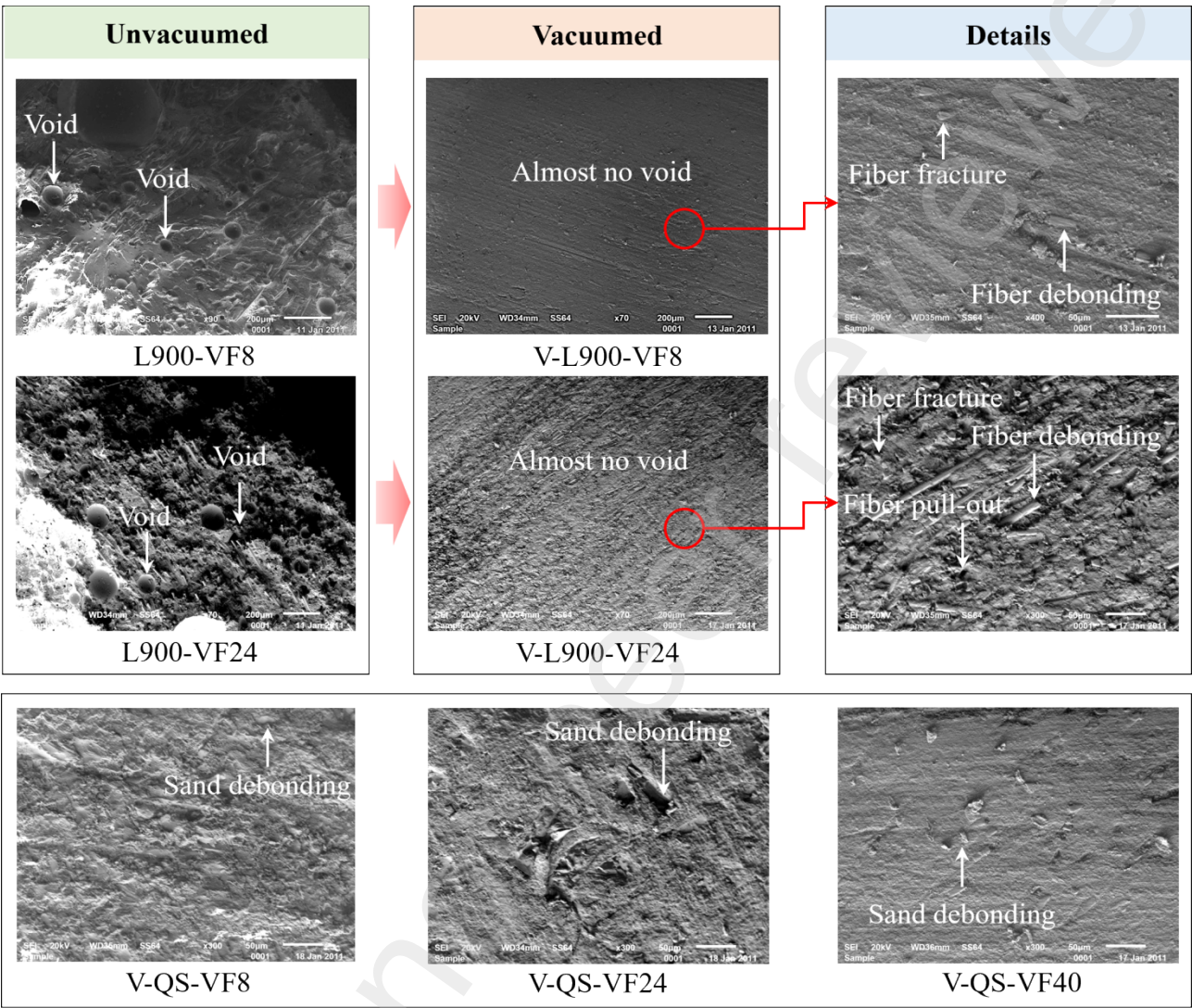


208 Fig. 5 Compressive strength and elastic modulus

209 2.2.4. Enhancement mechanism

210 The scanning electron microscope (SEM) micrographs in Fig. 6 show the inner structure of the
 211 modified resin before and after vacuuming, which was prepared by cutting treatment. For the
 212 unvacuumed resin (L900-VF8/24), numerous voids were observed, which increased with higher fiber
 213 VF. However, almost no voids were found after vacuuming (see V-L900-VF8/24). Voids reduce the
 214 effective stressed area and lead to stress nonuniformity and crack generation, thereby reducing the
 215 modification effect. Fiber debonding, pull-out, and fracture were observed, effectively increasing the
 216 energy absorption capacity of the modified resin. This enhancement effect increases with increasing
 217 fiber VF [29]. L900 exhibited superior enhancement compared to L150 due to its longer length and
 218 larger diameter, resulting in higher fiber bonding force and ultimate load. In contrast to fiber failure,
 219 sand debonding is prone to occur due to its small specific surface area; moreover, the small length-

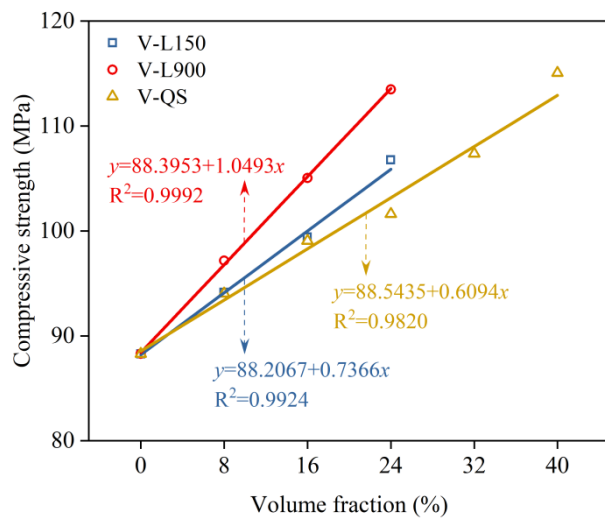
220 to-diameter ratio also resulted in insufficient crack control capacity. Consequently, the modified
 221 effect of sand is inferior to that of fibers.



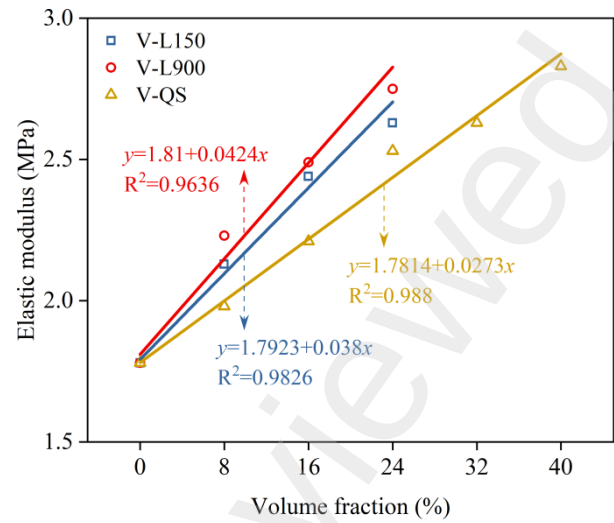
222
 223 Fig. 6 SEM micrograph of compressive specimens

224 **2.2.5. Prediction formula**

225 The linear formulas shown in Fig. 7 can effectively model the compressive strength and elastic
 226 modulus of the vacuumed modified resin. The fitting formulas consistently achieved an adjusted R^2
 227 value exceeding 0.98, indicating a high level of precision in the fitting process. The implementation
 228 of a variable-stiffness design for FRP CASs in a subsequent study is facilitated as a result.



(a) Compressive strength



(b) Elastic modulus

Fig. 7 Fitting formulas

3. Long-term compressive creep behavior of LTCs

3.1. Experimental program

3.1.1. Specimens

The compressive creep schemes are outlined in Table 6, in which S denotes the level of creep stress. The plain resin was utilized as a control group. Five levels of creep stress, namely 0.3, 0.5, 0.7, 0.8, and $0.9f_u$ were determined to assess the 1000-hour compressive creep behavior of the plain and modified resin. Three different VFs (8%, 16%, and 24%) were selected for SFMR and SSMR based on considerations of fluidity. The lever ratio was measured as 175:1, which was verified using a universal testing machine. Two identical specimens were prepared for each condition.

Table 6 Classification of compressive creep specimens

Number	Modified material	VF (%)	Stress level (f_u)	Lever ratio	Quantity
PR-S0.3/0.5/0.7/0.8/0.9	/	/	0.3/0.5/0.7/0.8/0.9	175:1	2/2/2/2/2
L900-VF8-S0.3/0.5/0.7/0.8/0.9	L900	8	0.3/0.5/0.7/0.8/0.9	175:1	2/2/2/2/2
L900-VF16-S0.3/0.5/0.7/0.8/0.9	L900	16	0.3/0.5/0.7/0.8/0.9	175:1	2/2/2/2/2
L900-VF24-S0.3/0.5/0.7/0.8/0.9	L900	24	0.3/0.5/0.7/0.8/0.9	175:1	2/2/2/2/2

QS-VF8-S0.3/0.5/0.7/0.8/0.9	Sand	8	0.3/0.5/0.7/0.8/0.9	175:1	2/2/2/2/2
QS-VF16-S0.3/0.5/0.7/0.8/0.9	Sand	16	0.3/0.5/0.7/0.8/0.9	175:1	2/2/2/2/2
QS-VF24-S0.3/0.5/0.7/0.8/0.9	Sand	24	0.3/0.5/0.7/0.8/0.9	175:1	2/2/2/2/2

3.1.2. Test setup

The compressive creep test was conducted by the standards GB/T 41061-2021 [32] and ASTM D2990-17 [33]. A compressive creep device, capable of simultaneously testing four specimens, was designed based on the lever principle as depicted in Fig. 8. Five identical devices were installed in the basement to minimize temperature fluctuations. The measured room temperature was approximately 19°C, which met the test requirements ($20 \pm 2^\circ\text{C}$). The original extension bar of the mechanical jack was replaced with a well-designed belt pulley to achieve continuous loading at a consistent lever ratio. The compressive load was applied by extending the jack piston and utilizing a reaction frame constructed with screws and steel plates. Small-diameter steel balls were utilized to adjust the weight accurately. The displacement of the piston, which was measured using a digital dial indicator, allows for the measurement of specimen deformation. Furthermore, a flat steel bar fixed onto the piston was utilized to transmit variations in displacement to the dial indicator.

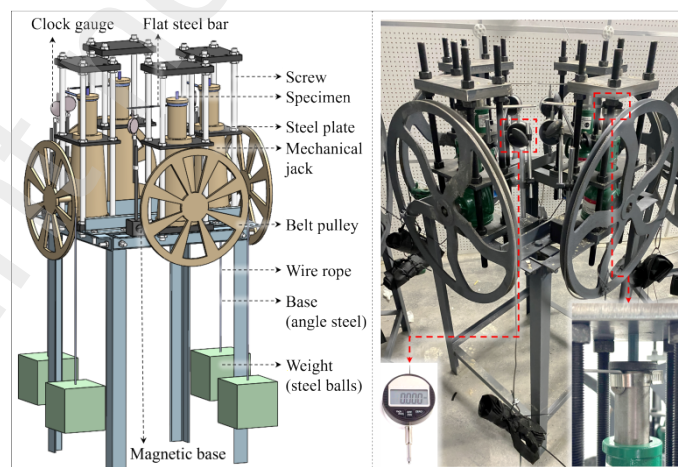
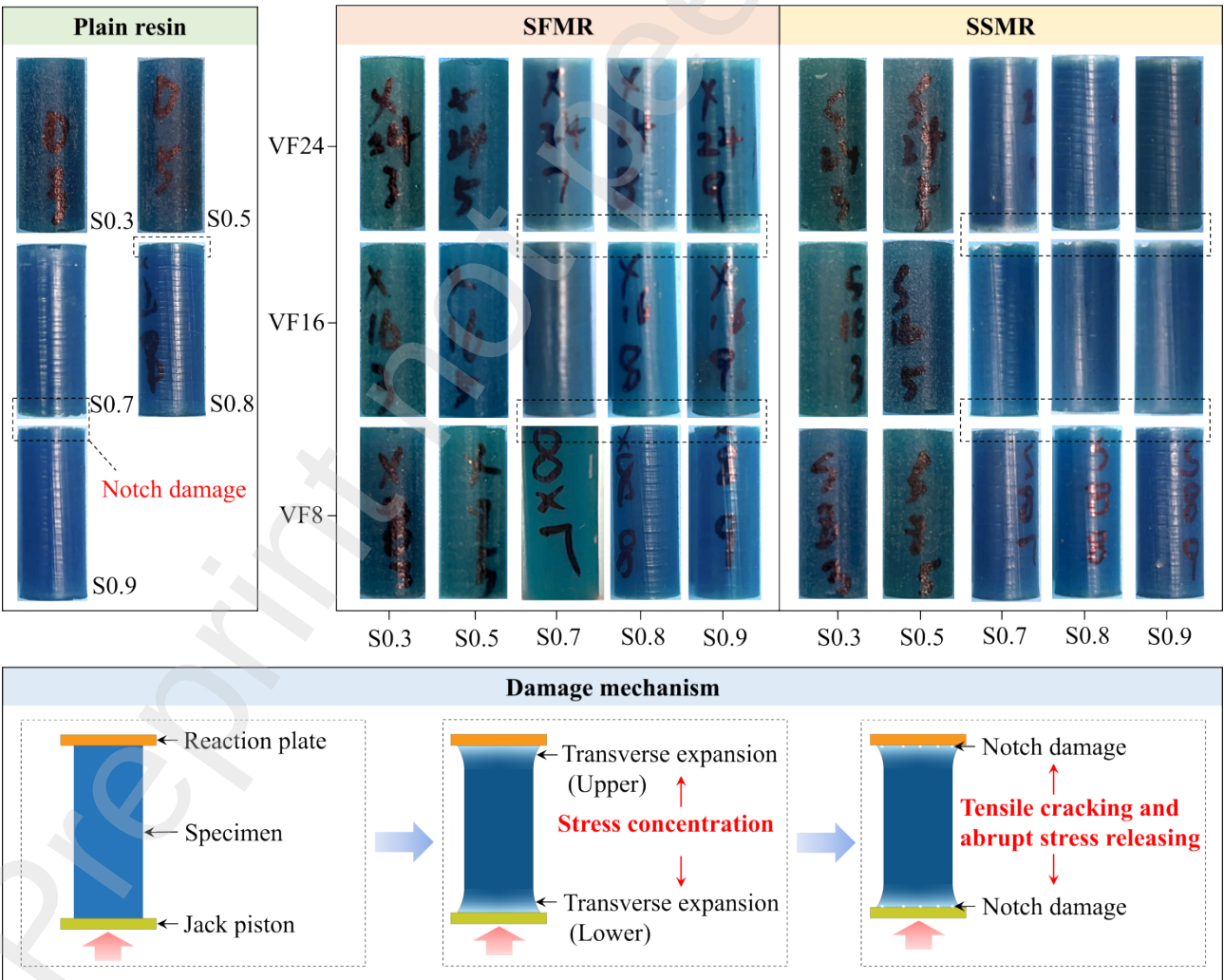


Fig. 8 Compressive creep test

256 **3.2. Creep results and discussions**

257 **3.2.1. Failure mode**

258 As illustrated in Fig. 9, whole buckling and shear cracks at the middle zone under variable stress
259 levels were not observed compared with Fig. 3. The local notch damages under high-stress levels of
260 0.7, 0.8, and $0.9f_u$ at the upper and lower stressed zones were found; meanwhile, the color of these
261 zones was lighter than middle position. This phenomenon can be explained by damage mechanism
262 analysis. The upper and lower zones of the specimens generated a transverse expansion deformation
263 due to compressive stress concentration, thereby resulting in a tensile cracking of the outside edge.
264 The modified resin at the cracking position experienced abrupt stress release, thereby developing into
265 notch damage.



266
267 **Fig. 9 Compressive creep failure modes mechanism**

268 **3.2.2. Creep strain-time curve**

269 The creep development of all specimens shown in Fig. 10 can be divided into two stages. The
270 first stage, characterized by rapid strain growth, was very short and was followed by a long-term
271 stable stage approximately 24 hours later. The first-stage creep strain increased gradually with higher
272 stress levels, while the second-stage creep strain showed minimal variations over time under different
273 stress levels. This indicated that compressive creep strains mainly occur during the first stage.
274 Therefore, precompression can be applied to relieve the first-stage creep deformation and ensure the
275 long-term behavior of FRP CASs. The growth rate of the 1000-hour compressive creep strain
276 nonlinearly increased with higher stress levels, indicating its greater sensitivity to high stress. SFMR
277 had the lowest 1000-hour creep strain, followed by SSMR and plain resin. Adding sand had a more
278 positive effect on reducing the creep strain of SSMR compared to SFMR.

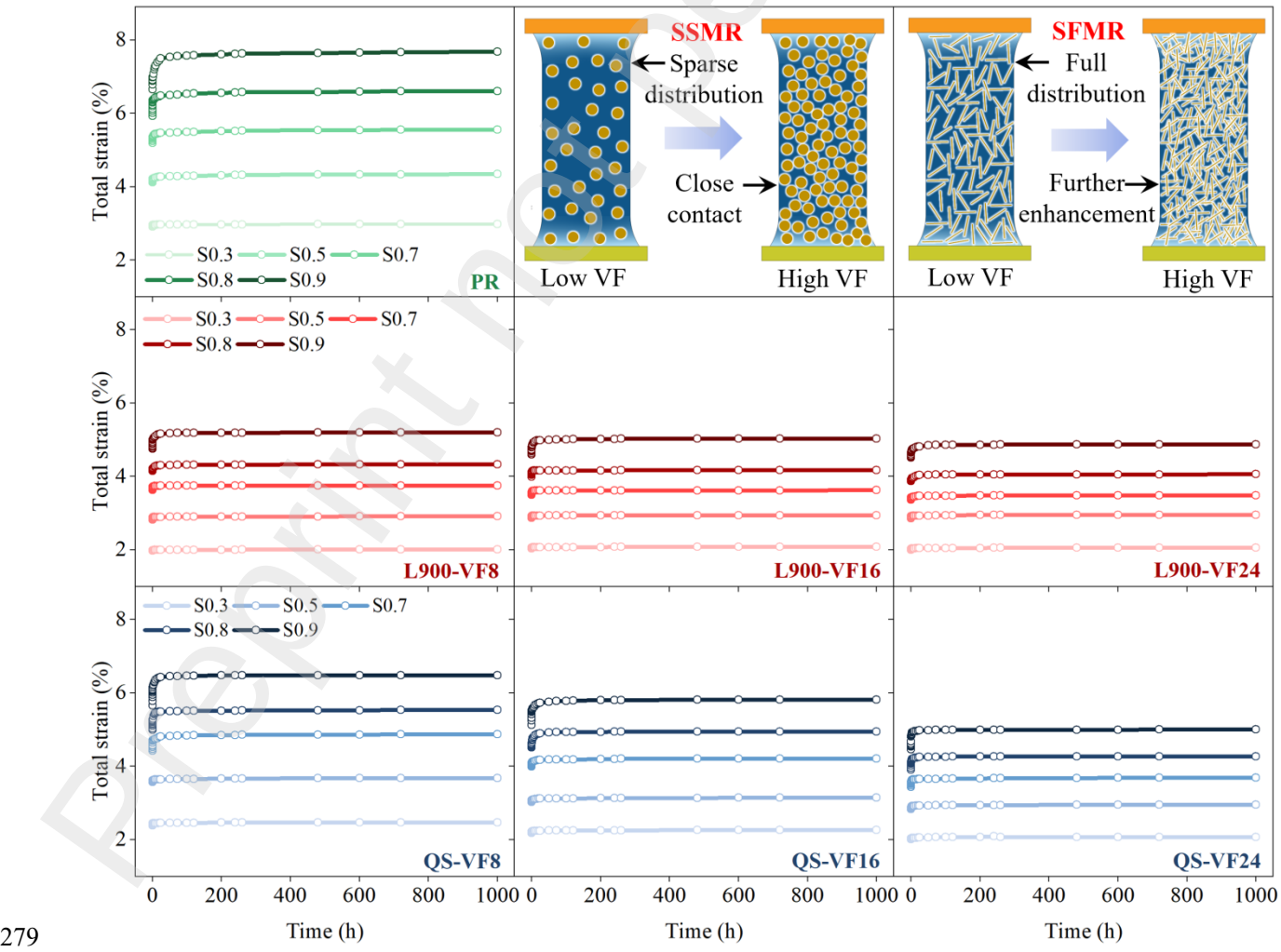


Fig. 10 Compressive creep strain-time curves

The aforementioned phenomenon can be explained by mechanism diagrams in Fig. 10. The specimen with a low sand VF in the mechanism diagrams showed a sparse distribution, enhancing the resin near sand locally. A high sand VF can significantly reduce resin content and achieve close contact distribution, forming a completely stressed skeleton that noticeably increases the creep behavior of the SSMR. When VF was the same, there were more fibers than sand due to their smaller fiber size. Additionally, fibers with a large length-to-diameter ratio can achieve full fiber distribution even with low VF, confining the resin within a three-dimensional web of fibers and controlling deformation. Thus, increasing VF enhanced fiber density and effectiveness. However, when fiber VF was high, aggregation occurred which reduced the enhancing effect of increasing fiber VF.

3.2.3. Creep coefficient

The 1000-hour creep coefficient represents the ratio of creep strain at a specific time to the initial elastic strain, which can be used to describe materials' creep deformation. Generally, the 1000-hour compressive creep coefficients of all specimens nonlinearly increased with higher stress levels, indicating its greater sensitivity to higher stress, as depicted in Fig. 11.

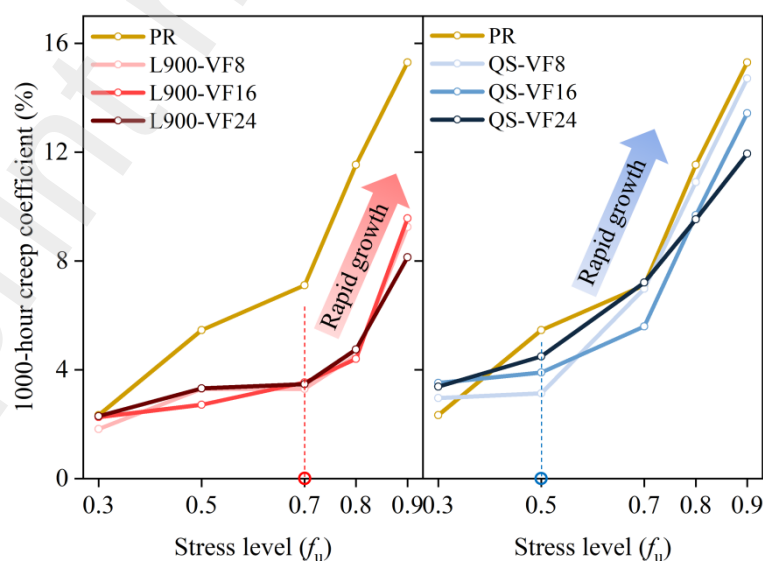


Fig. 11 1000-hour compressive creep coefficient

297 The creep coefficient of SFMR can be slightly affected by the fiber VF and was much lower
298 than that of the plain resin when stress levels were the same. Meanwhile, SFMR exhibited a rapid
299 growth in creep coefficient when the stress reached a level of $0.7f_u$. A similar trend can be observed
300 in SSMR, but SSMR experienced a rapid growth in creep coefficient when the stress level reached
301 $0.5f_u$. The creep coefficient of SFMR was much lower than that of SSMR when higher stress was
302 applied. In summary, adding fibers effectively reduced the creep coefficient of resin while it was
303 slightly affected by adding sand.

304 **3.2.4. Compressive strength retention**

305 The compressive strength retention of SFMR exhibited a similar variation pattern to SSMR, as
306 shown in Fig. 12. Generally, low-stress levels (0.3 and $0.5f_u$) had minimal impact on the strength
307 retention regardless of VFs of fibers and sand, with the strength retention still exceeding 90%. This
308 is because the modified resin tended to achieve stable and minor deformation without internal
309 cracking damage after undergoing 1000-hour creep. High stresses (0.7 , 0.8 and $0.9f_u$) had a more
310 significant effect on strength retention and were highly dependent on the VFs of fibers and sand.

311 Specifically, for SFMR, the strength retention exceeded 100% when the fiber VF ranged from 0
312 to 16%, mainly due to the straightening of the molecular chain in resin and fibers. However, when
313 the fiber VF reached 24%, the strength retention dropped below 80% due to interface debonding and
314 internal microcrack generation between resin and fibers. For SSMR, the strength retention can surpass
315 100% only when the sand VF is less than 8%, while it still exceeds 90% even with a higher sand VF.
316 The minimum strength retention for SFMR was found to be 74.8%, which was lower than that of
317 SSMR (93.7%). This difference arose because SFMR initially had a much higher strength compared
318 to SSMR, making it more susceptible to loss in strength. In summary, it is crucial to avoid excessive
319 stress levels and VFs to ensure sufficient strength retention.

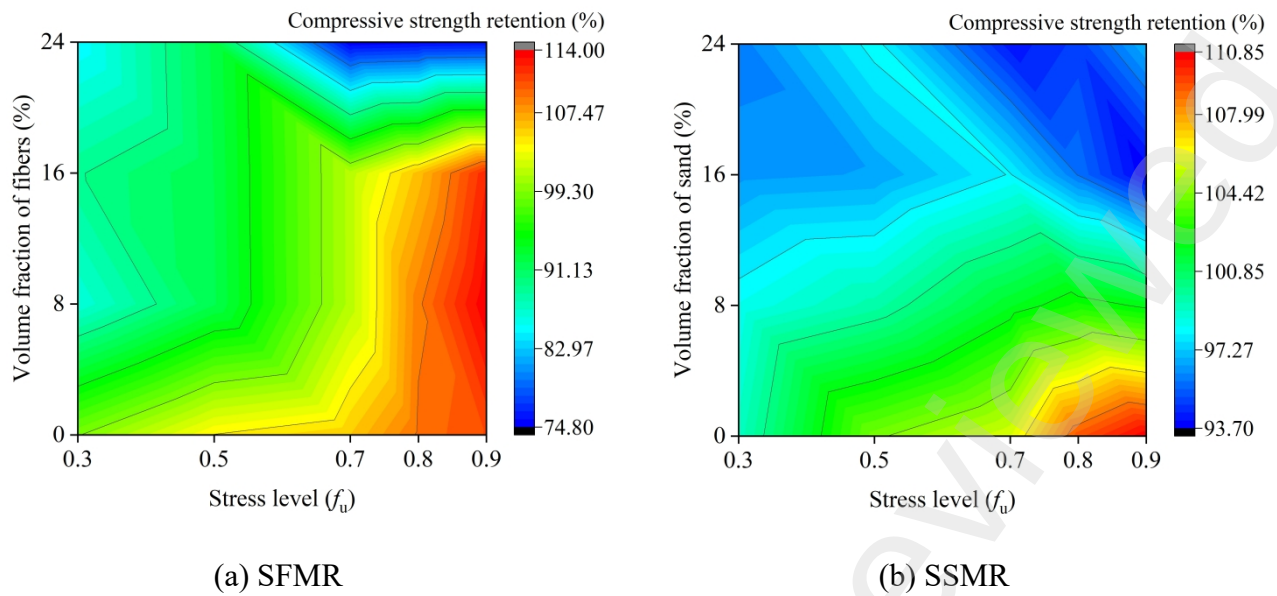


Fig. 12 Compressive strength retention

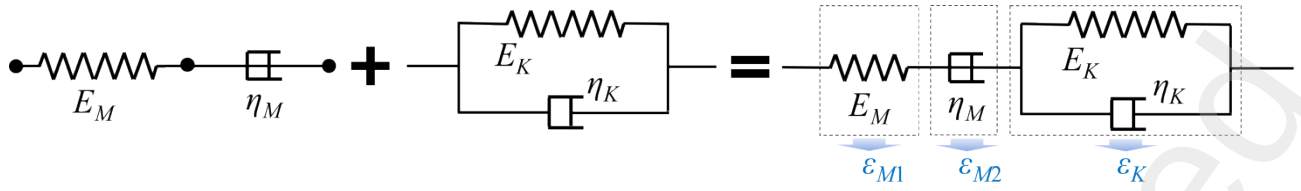
4. Prediction of compressive creep behavior

4.1. Creep prediction model

The cable-supported bridge has a design life exceeding 100 years, making the long-term creep behavior of the LTC a crucial factor in ensuring anchoring reliability. Consequently, typical models were utilized to predict the compressive creep behavior of resin-based LTCs for 10^6 hours (approximately 114 years). The Burges model (BM), Findley model (FM), logarithmic model (LM), and modified logarithmic model (MLM) in a previous study were employed to forecast the tensile creep performance of CFRP cables under various stress levels [27]. In this study, these models were also selected to evaluate the compressive creep behavior of plain resin, SFMR, and SSMR. The same letter represents the same meaning in subsequent analysis.

4.1.1 Burges model

The rheological Burgers model, as illustrated in Fig. 13, provides a more precise representation of viscoelastic behavior in composite materials by combining a Maxwell model and a Kelvin model in series. This model accurately captures both creep and relaxation behaviors [34]. Elastic behavior is represented by a spring, while viscous behavior is denoted by a dashpot.



(a) Maxwell model

(b) Kelvin model

(c) Burgers model

Fig. 13 Rheological models

A four-parameter Burgers model can accurately depict the correlation among stress, strain, and time, with its strain being categorized into three distinct parts, as expressed in Eq. (3).

$$\varepsilon(t) = \varepsilon_{M1} + \varepsilon_{M2} + \varepsilon_K \quad (3)$$

where $\varepsilon(t)$ represents the total strain, ε_{M1} denotes the instantaneous elastic strain calculated by the spring element with an elastic modulus of E_M , ε_{M2} is the viscous strain calculated by the dashpot with a viscosity coefficient of η_M , and ε_K signifies the delayed elastic strain calculated by the Kelvin model. These relationships are represented by Eqs. (4–6):

$$\varepsilon_{M1} = \frac{\sigma_0}{E_M} \quad (4)$$

$$\varepsilon_{M2} = \frac{\sigma_0}{\eta_M} t \quad (5)$$

$$\varepsilon_K = \frac{\sigma_0}{E_K} \left(1 - e^{-\frac{E_K}{\eta_K} t} \right) \quad (6)$$

where σ_0 denotes the initial creep stress and t is the creep time. Based on the equations mentioned above, the total strain can be redefined as Eq. (7):

$$\varepsilon(t) = \sigma_0 \left[\frac{1}{E_M} + \frac{t}{\eta_M} + \frac{1}{E_K} \left(1 - e^{-\frac{E_K}{\eta_K} t} \right) \right] \quad (7)$$

Equation (7) can be further simplified to Eq. (8) by using four uncertain coefficients.

$$\varepsilon(t) = a + bt + c(1 - e^{-dt}) \quad (8)$$

4.1.2. Findley model

The Findley model [35], proposed based on empirical formulas, features an exponential form

358 for predicting the creep of viscoelastic materials. The total strain can be expressed as Eq. (9):

359
$$\varepsilon(t) = \varepsilon_0 + mt^n \quad (9)$$

360 where ε_0 represents the initial elastic strain, m is a coefficient associated with the stress and
361 temperature, and n is a material constant related to the stress.

362 4.1.3. Logarithmic model

363 Research has verified that the tensile creep strain of FRP materials exhibits a linear relationship
364 with the logarithm of time [12]. The total strain can be expressed as Eq. (10):

365
$$\varepsilon(t) = \varepsilon_0 + a \ln t \quad (10)$$

366 where a denotes a coefficient related to the stress and material.

367 4.1.4. Modified logarithmic model

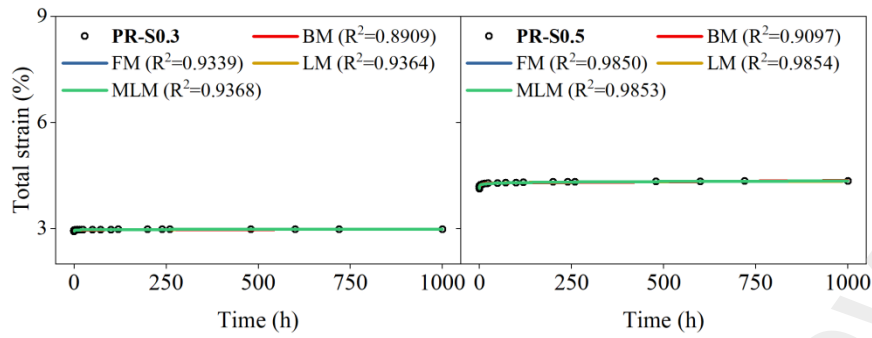
368 The modified logarithmic model has an additional linear expression compared with the
369 logarithmic model. It has been demonstrated that the modified logarithmic model exhibits superior
370 accuracy in fitting the creep behavior of CFRP when compared to various other models [36]. The
371 total strain can be expressed as Eq. (11):

372
$$\varepsilon(t) = \varepsilon_0 + a \ln t + bt \quad (11)$$

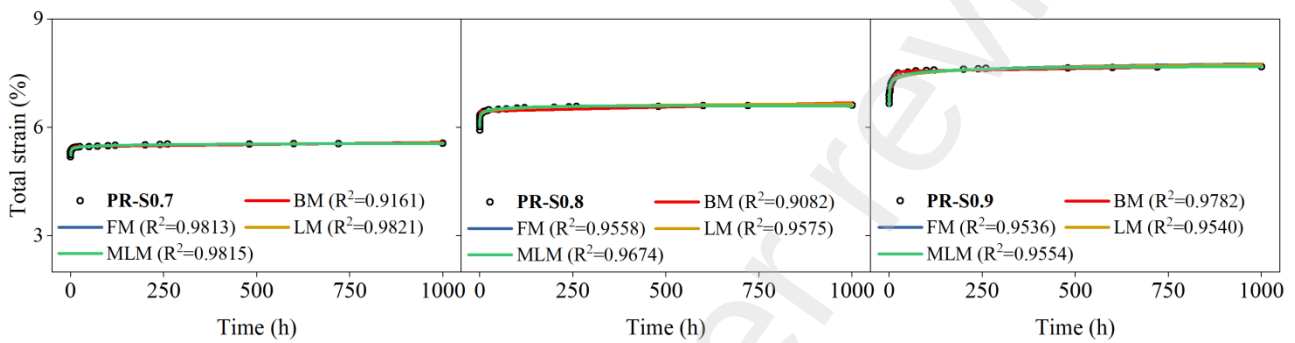
373 where b is a coefficient related to the stress and material.

374 4.2. Fitting of 1000-hour creep strain-time curves

375 The 1000-hour compressive creep strain-time curves of the plain resin under different stress
376 levels were described in Fig. 14. The adjusted R^2 was utilized to assess the accuracy of predictions,
377 allowing for the exclusion of variable quantity effects. Except for PR-S0.3 fitted by the Burges model
378 ($R^2=0.8909$), the R^2 values of all prediction models exceeded 0.90, indicating that these models can
379 reliably describe the 1000-hour creep behaviors of the plain resin. On this basis, the aforementioned
380 models can be further used to forecast million-hour creep behavior.



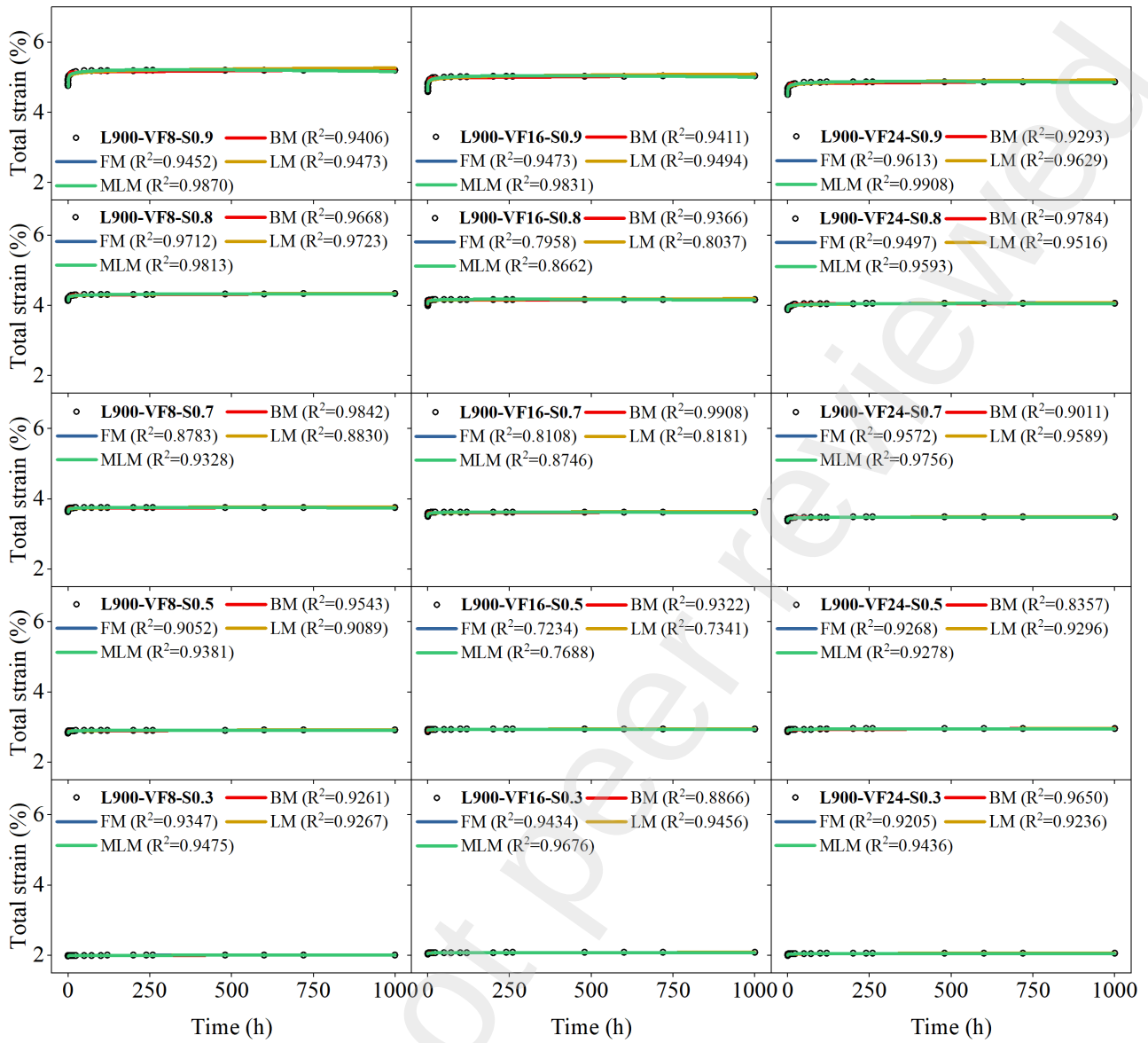
(a) Stress levels = 0.3 and $0.5f_u$



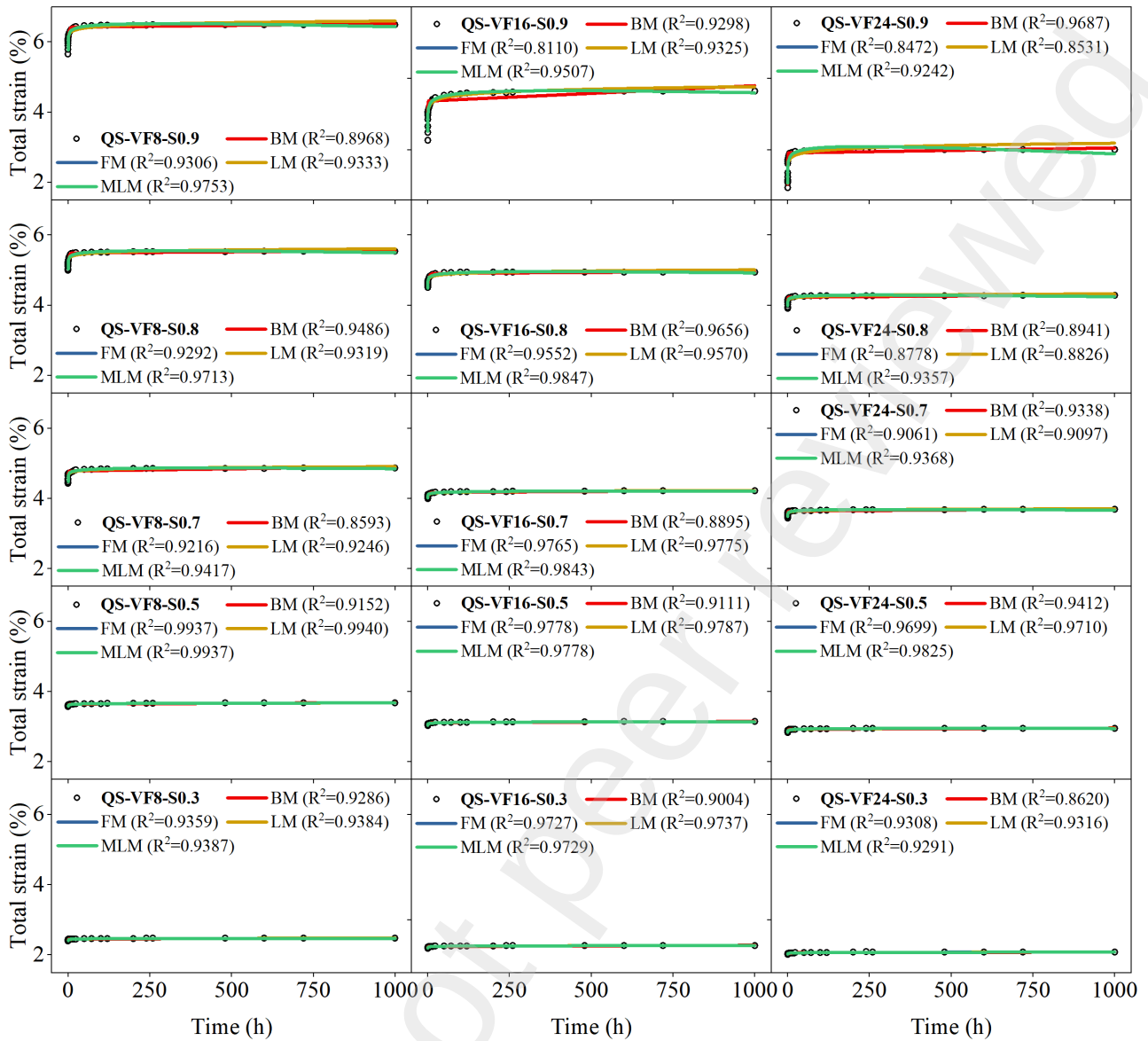
(b) Stress levels = 0.7, 0.8 and $0.9f_u$

Fig. 14 Fitted curves of plain resin

Similarly, the aforementioned four models were utilized to forecast the 1000-hour compressive creep strains of SFMR and SSMR, as depicted in Fig. 15. For the majority of experimental specimens, the R^2 values of the fitted curves generated by different prediction models consistently surpassed 0.90, indicating that these models exhibited a high degree of applicability for various stress levels and material types. Although some R^2 values were lower than 0.90, the corresponding fitted curves still effectively captured the second-stage creep deformation. In other words, the decrease in R^2 values primarily resulted from variations in first-stage creep deformation behavior. The significance and relevance lie in understanding that second-stage creep deformation is more crucial as it reflects the ultimate deformation.



(a) SFMR



(b) SSMR

Fig. 15 Fitted curves of modified resin

4.3. Prediction of million-hour total strains and creep coefficients

The total strains and creep coefficients of the plain resin after one million hours are summarized in Table 7. In contrast to the 1000-hour prediction results, there were significant discrepancies among the million-hour predictions from different models. Specifically, the Burges model overestimated the long-term creep development due to its inclusion of a linear expression dependent on time, which could lead to infinite strain with increasing time. Additionally, the modified logarithmic model tended to underestimate long-term creep as it can also be affected by a linear expression dependent on time.

When the stress levels were 0.3, 0.5, 0.7, 0.8, and $0.9f_u$, the corresponding million-hour creep coefficients predicted by the Findley model were determined as 3.4%, 8.8%, 11.0%, 18.3%, and 28.5% respectively. Moreover, it was observed that the logarithmic model demonstrated a comparable predictive capability to that of the Findley model. Notably, the total strain corresponding to the maximum load in the static test results was 6.9%, indicating that the maximum stress level should be controlled below $0.7f_u$. For achieving a creep coefficient within a range of approximately 10% in plain resin, it is recommended to limit the maximum stress level to around $0.5f_u$. In summary, both the Findley model and logarithmic model can be considered reliable tools for reasonably forecasting million-hour compressive creep behavior in plain resin.

Table 7 Predicted million-hour total strains and creep coefficients of the plain resin (10^6 h)

Number	Total strain (%)					Creep coefficient (%)				
	1000 hour	BM	FM	LM	MLM	1000 hour	BM	FM	LM	MLM
PR-S0.3	3.0	29.4	3.0	3.0	-1.6	2.3	907.1	3.4	3.4	-154.0
PR-S0.5	4.3	75.8	4.5	4.5	11.4	5.5	1740.9	8.8	8.5	177.2
PR-S0.7	5.6	121.4	5.8	5.8	-1.2	7.11	2240.0	11.0	11.0	-123.8
PR-S0.8	6.6	239.8	7.0	7.0	-96.0	11.5	3945.0	18.3	18.3	-1718.6
PR-S0.9	7.7	164.8	8.6	8.4	-88.0	15.3	2374.8	28.5	26.2	-1422.1

The prediction trends of SFMR and SSMR using different models were consistent with those of the plain resin, as summarized in Table 8 and Table 9. Specifically, all four models effectively captured the creep behaviors over 1000 hours, while only the Findley model and logarithmic model accurately reflected the creep behaviors over million hours. The total strains of V-L900-V8/16/24 corresponding to the maximum load in the static test results were 5.1%, 5.2%, and 5.5% respectively, indicating that the maximum stress level should be set to about $0.9f_u$. Additionally, the total strain of SFMR after one million hours under various stress levels did not exceed 6%, with corresponding creep coefficients within 8%.

425 In contrast, SSMR exhibited significantly higher million-hour creep coefficients ranging from
 426 4.9% to 24.9%, surpassing those observed in SFMR with identical VFs. The total strains of V-QS-
 427 V8/16/24 corresponding to the maximum load in the static test results were 6.1%, 4.9%, and 5.1%
 428 respectively, indicating that the maximum stress level should be controlled below $0.5f_u$. In conclusion,
 429 for achieving a creep coefficient within a range of approximately 10%, it is recommended to limit the
 430 maximum stress level to around $0.8f_u$ for SFMR and $0.5f_u$ for SSMR; both the Findley model and
 431 logarithmic model can reliably predict the long-term creep behaviors of SFMR and SSMR. The use
 432 of SFMR-based LTC can ensure satisfactory long-term performance for FRP CAS.

433 Table 8 Predicted million-hour total strains of SFMR and SSMR (10^6 h)

Number	SFMR (%)					SSMR (%)				
	1000 hour	BM	FM	LM	MLM	1000 hour	BM	FM	LM	MLM
VF8-S0.3	2.0	867.7	2.0	2.0	9.7	2.5	35.7	2.5	2.5	-3.7
VF8-S0.5	2.9	747.2	3.0	3.0	-22.5	3.7	47.2	3.7	3.7	4.0
VF8-S0.7	3.8	546.8	3.9	3.9	-48.4	4.9	130.0	5.2	5.2	-87.5
VF8-S0.8	4.3	941.7	4.5	4.5	-28.4	5.5	66.4	5.9	5.9	-174.7
VF8-S0.9	5.2	1563.6	5.6	5.6	-156.4	6.5	103.2	7.1	7.1	-255.1
VF16-S0.3	2.1	630.0	2.1	2.1	-8.1	2.3	38.9	2.3	2.3	0.6
VF16-S0.5	2.9	545.6	3.0	3.0	-16.2	3.1	36.7	3.2	3.2	2.9
VF16-S0.7	3.6	468.7	3.7	3.7	-46.9	4.2	57.6	4.4	4.4	-24.6
VF16-S0.8	4.2	649.9	4.3	4.3	-58.8	4.9	56.9	5.3	5.3	-130.3
VF16-S0.9	5.0	1614.5	5.4	5.4	-130.4	5.8	220.6	6.2	6.2	-131.3
VF24-S0.3	2.1	1003.9	2.1	2.1	-14.3	2.1	9.9	2.1	2.1	0.4
VF24-S0.5	3.0	1461.0	3.0	3.0	-0.8	3.0	37.7	3.0	3.0	-18.3
VF24-S0.7	3.5	393.2	3.6	3.6	-20.8	3.7	61.7	3.8	3.8	-55.9
VF24-S0.8	4.1	373.2	4.2	4.2	-30.1	4.3	67.9	4.5	4.5	-121.9
VF24-S0.9	4.9	1753.7	5.2	5.2	-107.6	5.0	75.0	5.4	5.4	-233.5

434 Table 9 Predicted million-hour creep coefficients of SFMR and SSMR (10^6 h)

Number	SFMR (%)					SSMR (%)				
	1000 hour	BM	FM	LM	MLM	1000 hour	BM	FM	LM	MLM
VF8-S0.3	1.8	867.7	3.5	2.6	389.7	3.0	1384.2	5.0	5.0	-254.0

VF8-S0.5	3.3	747.2	5.5	5.5	-896.6	3.1	1222.9	4.9	4.9	12.7
VF8-S0.7	3.3	546.8	6.1	6.1	-1432.4	7.0	2835.8	16.4	16.4	-2075.3
VF8-S0.8	4.5	941.7	7.9	7.9	-785.7	10.9	1230.0	18.9	18.9	-3599.2
VF8-S0.9	9.3	1563.6	17.0	17.0	-3388.4	14.7	1724.5	24.9	24.9	-4611.4
VF16-S0.3	2.3	630.0	3.7	3.7	-497.1	3.5	1673.0	5.6	5.6	-73.2
VF16-S0.5	2.7	545.6	4.0	4.0	-667.2	3.9	1110.8	6.0	6.0	-4.4
VF16-S0.7	3.5	468.7	6.0	6.0	-1438.7	5.6	1342.8	9.1	9.1	-716.9
VF16-S0.8	4.4	649.9	7.0	7.0	-1571.5	9.7	1162.7	17.7	17.7	-2990.5
VF16-S0.9	9.6	1614.5	16.7	16.7	-2937.9	13.4	4198.8	21.6	21.6	-2659.8
VF24-S0.3	2.3	1003.9	5.6	5.6	-814.8	3.4	390.7	6.2	5.5	-79.4
VF24-S0.5	3.3	1461.0	5.0	5.0	-129.0	4.5	1233.9	7.4	7.4	-746.3
VF24-S0.7	3.5	393.2	5.8	5.8	-717.6	7.2	1693.9	11.5	11.5	-1726.2
VF24-S0.8	4.8	373.2	8.8	8.7	-877.7	9.5	1639.1	15.7	15.7	-3222.0
VF24-S0.9	8.1	1753.7	14.8	14.8	-2486.3	12.0	1579.1	20.8	20.8	-5325.4

5. Conclusions

The static compressive property and long-term creep behavior of LTCs were evaluated experimentally and theoretically. The key findings are presented as follows.

(1) Vacuuming increased compressive strength by 18.0% to 113.3% and elastic modulus by 22.5% to 48.0%; these properties of SFMR and SSMR show a linear relationship with an adjusted R^2 value exceeding 0.98.

(2) High-stress levels ($\geq 0.7f_u$) can cause notch damage in the upper and lower stressed zones.

All specimens exhibited a two-stage law for creep development. Adding sand has a more positive impact on reducing the creep strain of SSMR compared to SFMR.

(3) Adding fibers reduced the 1000-hour creep coefficient while adding sand had a slight effect; the creep coefficients of SFMR and SSMR rapidly increased when stress levels reached $0.7f_u$ and $0.5f_u$ respectively. Excessive stresses and VFs reduced strength retention after creep.

(4) Both the Findley model and logarithmic model can reasonably forecast million-hour creep behavior. To achieve a creep coefficient within 10%, the maximum stress levels for plain

resin, SFMR, and SSMR should be set at $0.5f_u$, $0.8f_u$, and $0.5f_u$ respectively.

Credit authorship contribution statement

Jingyang Zhou: Methodology, Formal analysis, Investigation, Data curation, Writing-Original Draft, Funding acquisition, Review & Editing. **Haorui Xiang:** Methodology, Formal analysis, Investigation. **Xin Wang:** Conceptualization, Review & Editing, Funding acquisition, Project administration. **Leizhi Zhang:** Formal analysis, Investigation. **Xinquan Chang:** Review & Editing. **Qiwei Xu:** Investigation & Data curation. **Zhishen Wu:** Supervision.

Declaration of Interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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